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Genetic Network Shaping Kenyon Cell Identity and Function in *Drosophila* Mushroom Bodies

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eLife Assessment

This **fundamental** study uses the *Drosophila* mushroom body as a model to understand the molecular machinery that controls the temporal specification of neuronal cell types. With **convincing** experimental evidence, the authors make the finding that the Pipsqueak domain-containing transcription factor Eip93F plays a central role in specifying a later-born neuronal subtype while repressing gene expression programs for earlier subtypes.

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Abstract

Revealing the molecular mechanisms underlying neuronal specification and acquisition of specific functions is key to understanding how the nervous system is constructed. In the *Drosophila* brain, Kenyon cells (KCs) are sequentially generated to assemble the backbone of the mushroom body (MB). Broad-complex, tramtrack and bric-à-brac zinc finger transcription factors (BTBzf TFs) specify early-born KCs, whereas the essential TFs for specifying late-born KCs remain unidentified. Here, we report that Pipsqueak domain-containing TF *Eip93F* promotes the identity of late-born KCs by reciprocally regulating gene expression in main KC types. Moreover, *Eip93F* not only regulates the expression of calcium channel *Ca-α1T* in late-born KCs to functionally control animal behavior, but it also forms a genetic network with *BTBzf* TFs to specify the identities of main KC types. Our study provides crucial information linking KC-type diversification to unique function acquisition in the adult MB.

Introduction

The nervous system contains a diverse ensemble of neuron types, which are generated during development in appropriate numbers under tightly controlled processes. A central question in developmental neurobiology concerns how diverse neuron types acquire unique cell identities with characteristic morphologies, distinct gene expression profiles and specific functionalities. Kenyon cells (KCs) are the intrinsic neurons in the *Drosophila* mushroom body (MB) [1] and can serve as an excellent model system for investigations into the molecular mechanisms of neuron-type diversification. Around 2,000 KCs generated by four neuroblasts are grouped into three sequentially born types, including γ , α'/β' and α/β neurons [1]. Functionally, these main KC types play different roles in short-term memory, acquisition, stabilization, and retrieval of memory [2, 3]. Morphologically, both α/β and α'/β' neurons have axons with two branches that project dorsally and medially into respective α and β , or α' and β' lobes, whereas γ neurons send out single axon branches that project medially into the γ lobe [1]. These three types of KCs can be also distinguished by differentially expressed marker genes. For instance, broad-complex, tramtrack and bric-à-brac zinc finger transcription factor (BTBzf TF) Abrupt (Ab) is specifically expressed in γ neurons [4–6], whereas cell adhesion molecule Fasciclin II and Rho guanine nucleotide exchange factor Trio are expressed in different subsets of KC types [4]. Therefore, specifying KC types with proper cell numbers, unique cell identities and distinct functions is a key aspect of MB formation.

Previous studies have revealed that the other BTBzF TF, Chronologically inappropriate morphogenesis (Chinmo), exhibits a graded expression pattern in KCs and plays a crucial role in diversifying main KC types [7]. Chinmo is highly expressed in early-born γ neurons, but its expression level is low in later-born α'/β' and absent in α/β neurons [7]. Notably, Chinmo controls the expression of the third BTBzF TF, Maternal gene required for meiosis (Mamo), functioning to specify the identities of main KC types at the larval stage via a fine-tuning process. In particular, a high expression level of Chinmo inhibits Mamo expression in γ neurons, whereas low Chinmo expression promotes Mamo production in α'/β' neurons [4]. As a gradual reduction of Chinmo expression occurs during development, Mamo, adding to its function in α'/β' neurons, takes over Chinmo's role of regulating the differentiation of γ neurons at the pupal stage [4, 5, 7]. In the absence of *chinmo* and *mamo*, molecular and morphological characteristics of γ and α'/β' neurons are shifted towards those of α/β neurons, constituting a cell identity transformation [4, 5, 7]. Therefore, the late-born α/β neural identity is hypothesized to be a default status upon the loss of the neural identity determinants for early-born KCs [4, 7]. However, it is also possible that key specification regulators of α/β neurons exist but have not yet been identified.

In this study, we leverage RNA-seq databases on KCs to identify type-specific markers as readouts for the cell identity of γ , α'/β' and α/β neurons [8, 9]. By doing so, we identified a phylogenetically conserved Pipsqueak domain-containing TF, Ecdysone-induced protein 93F (Eip93F/E93), with preferential expression in α/β neurons. Loss-of-function of *E93* not only downregulated α/β -specific gene expression, including a subunit of T-type like voltage gated calcium channel *Ca- α 1T*, but it also upregulated γ -specific *Ab* in late-born KCs, implying that an identity shift towards early-born KCs had occurred. Intriguingly, RNAi knockdown of *E93* or *Ca- α 1T* in α/β neurons further compromised animal behaviors, including foraging-related and night-time activities. In contrast, *E93* overexpression precociously turned on *Ca- α 1T* expression in early-born KCs at the expense of abolishing expression of early-born KC markers, such as *Ab* and *Mamo*. Notably, *E93* was upregulated in early-born KCs in the absence of *chinmo* and *mamo* but diminished in late-born KCs upon *Ab* overexpression. Taken together, our results suggest that a hierarchical genetic network among *chinmo*, *mamo*, *E93* and *ab* with potential feedback loops controls the identity and function of main KC types during the construction of functional MBs.

Results

Identification of KC-type specific markers

By leveraging information from published RNA-seq studies showing preferentially expressed genes in adult KC types [8, 9], we sought to identify a collection of KC-type specific marker lines (available at stock centers) with GFP transgenes at genes of interest [10–12]. The results of this screen for KC marker GFP lines are depicted in S1 Fig and Supplemental table 1; highlights of certain KC-type specific marker lines are described below. First, Abrupt (*Ab*)-GFP, a BAC clone-engineered GFP line, can be used to replace an excellent *Ab* antibody (generated by Dr. Crews laboratory but no longer available) for labeling γ neurons [4–6] (Fig 1A–B and S2 Fig). In addition, Lachesin (*Lac*; an Ig superfamily protein[13])-FSVS, a GFP-trapping line, expresses GFP-fused *Lac* specifically in α'/β' neurons starting from the early pupal stage (Fig 1C–D and S3A Fig). Notably, we also identified two other GFP-trapping lines, Ecdysone-induced protein 93F (*E93*)-GFSTF and Calcium channel protein α 1 subunit T (*Ca- α 1T*)-GFSTF, that express GFP-fused proteins enriched in α/β neurons. Enrichment in these neurons is evidenced by patterns of complementary expression to *Trio*, a marker of γ and α'/β' neurons [4] (Fig 1E–H). Consistent with the notion that generation of most of α/β neurons occurs at the pupal stage [1], *E93*-GFSTF and *Ca- α 1T*-GFSTF were not detectable in KCs at the wandering larval (WL) stage (Fig 1E, 1G and S3B Fig). With this set of useful reagents, we set out to further explore the molecular mechanisms underlying cell identity specification of main KC types.

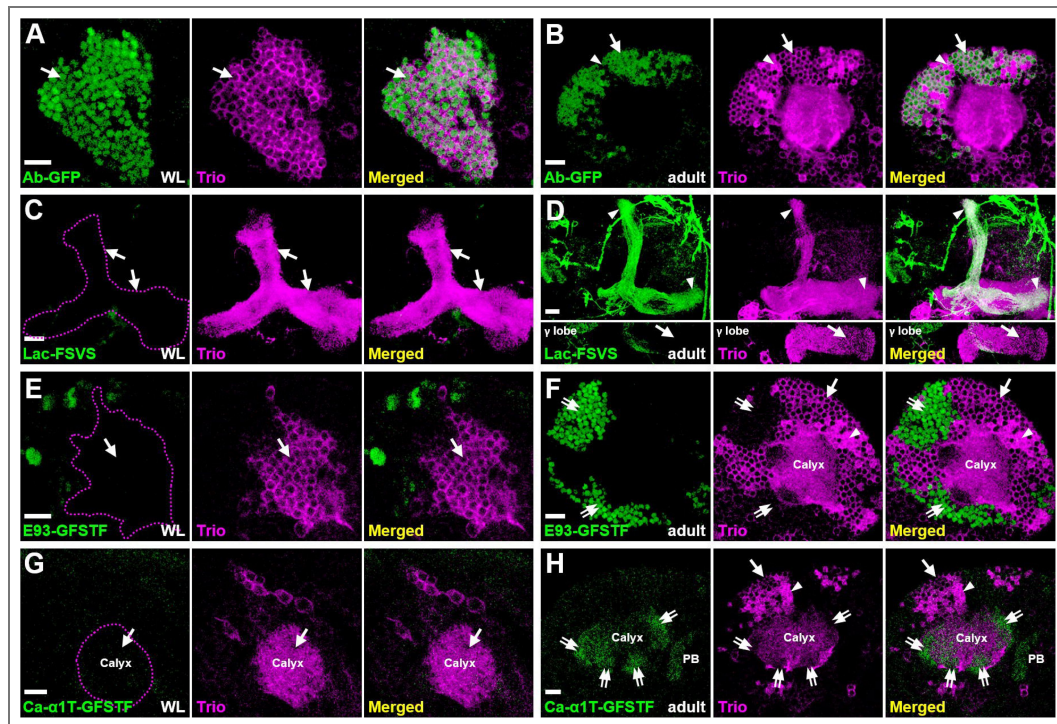


Fig 1. Expression patterns of main KC type markers

(A-H) KC type-specific GFP markers (green) were counter-stained with Trio (magenta) to reveal expression patterns at the wandering larval (WL) (A, C, E and G) and adult (B, D, F and H) stages. (A-B) Ab-GFP was primarily expressed in cell bodies of γ neurons at both WL and adult stages. The Trio signal indicates locations of γ neurons for staining observed only in cytosol (arrows) and α'/β' neurons for staining in the entire cell (arrowheads). (C-D) Lac-FSVS expression was enriched in α' and β' lobes (arrowheads) of adult but not WL stage animals. The single section in the bottom panels of Fig. 1D reveal the lack of Lac-FSVS expression in the γ lobe. (E-H) E93-GFSTF and Ca- α 1T-GFSTF were preferentially expressed in respective cell bodies and dendrites (the calyx) of α/β neurons (double-arrows) at adult but not WL stage animals. In addition to calyx expression, Ca- α 1T-GFSTF was also seen in the protocerebral bridge (PB) of adult brains. Genotypes shown in all figures are summarized in Supplementary Table 2. Scale bar: 10 μ m.

Loss-of-function of *E93* suppresses the α/β -neural identity and has behavioral consequences

Since *E93*, a Pipsqueak-domain containing TF, functions crucially in various biological processes [14–17] and *E93*-GFSTF was preferentially expressed in α/β neurons (Fig 1F), we sought to investigate whether *E93* acts as a critical regulator for specifying the α/β -neural identity. First, we found that the expression of a α/β -specific marker, *Ca- α 1T*-GFSTF, was diminished in the context of *E93* mutation (*E93* ^{Δ 11}) and in a line with *E93* knockdown due to overexpression of *E93* RNAi by a pan-KC driver, GAL4-OK107 [1] (Fig 2A–B and S4–S5 Fig). In contrast, the same RNAi knockdown elicited no discernible effects on the expression levels of *Trio* and *Lac-FSVS* in γ and α'/β' neurons (Fig 2A–B and S6 Fig). In addition to the observed reduction of *Ca- α 1T*-GFSTF, *E93* knockdown in KCs substantially compromised the expression of another α/β -specific marker, *myr::GFP* driven by 44E04-LexA [5] (Fig 2C–D). Although these results seemed to indicate the loss of α/β neural identity due to loss of *E93* function, the findings also raised a possibility that the reduction of *Ca- α 1T*-GFSTF and 44E04-LexA might simply be due to the absence of late-born KCs. However, this explanation was ruled out by the observation of relatively intact morphology of α and β lobes when we examined the expression of a third α/β -specific marker (*myr::GFP* driven by 70F05-LexA) upon *E93* knockdown in KCs [5] (Fig 2E–F). Moreover, compared to wild-type samples, the cell number of 70F05-LexA-positive neurons was not significantly altered in KCs when *E93* was knocked down (Fig 2G–H and S7 Fig). Taking advantage of 70F05-LexA as a putative α/β -neural marker, we further found that more than half of the 70F05-LexA-positive neurons ectopically expressed γ -specific *Ab-GFP* in *E93* knockdown samples (Fig 2G–H and S7 Fig), implying that α/β neurons had transformed into γ -like neurons. These results taken together suggested that *E93*, despite of unable to direct axon morphogenesis, is required for specifying KCs towards the cell identity of α/β neurons by regulating the expression of certain α/β -specific genes.

Since *Ca- α 1T* encodes a subunit of a T-type like voltage gated calcium channel, which regulates sleep behavior [18], we wondered whether the loss of *E93* and *Ca- α 1T* in α/β neurons could cause behavioral defects. To examine the behavior, we utilized a monitoring system to film and analyze the activities of individual flies (S8A–B Fig). The animal's speed of locomotion usually peaks around the day-to-night shift and gradually becomes reduced at night. In control samples (wild-type, RNAi-only, and GAL4-only lines), it took around eight hours for the moving speed to reach a minimum during the night period (Fig 2I). Interestingly, night-time activity was significantly impaired (with a sharp reduction of moving speed to a minimal value at around two to four hours into the night period) when *Ca- α 1T* and *E93* were knocked down using α/β neural drivers, 13F02-AD/70F05-DBD and *c739*-GAL4 [19] (Fig 2I and S8C–F Fig). Although the overall moving speed was lower in samples of *E93* knockdown driven by *c739*-GAL4 than in samples of other genotypes, the wings of these *E93* knockdown flies appeared curly and aberrant, which might cause the movement defect. In addition to the effects on night-time activity, RNAi knockdown of *Ca- α 1T* and *E93* seemed to compromise the foraging-related behavior (S8C Fig). Compared to control flies, *Ca- α 1T*-deficient animals (especially for RNAi knockdown by *c739*-GAL4) tended to explore regions without fly food during the day-time period (Fig 2I and S8F Fig). Taken together, these results suggested that *Ca- α 1T* (whose expression is regulated by *E93*) acts in α/β neurons to potentially control animal behavior.

E93 overexpression promotes α/β neural traits in early-born KCs

We next asked if *E93* indeed plays a crucial role in specifying α/β neural identity, i.e., does gain of *E93* function transform the cell identity of other KC types to α/β neurons? In contrast to the undetectable *Ca- α 1T*-GFSTF expression in wild-type with two main KC types, γ and α'/β' neurons, at the WL stage, *Ca- α 1T*-GFSTF expression was precociously upregulated in KCs when *E93* was overexpressed by GAL4-OK107 (Fig 3A–B). Similarly, a portion of putative γ neurons exhibited α/β -neural like features (*myr::GFP* driven by R70F05-LexA) with axonal defects when *E93* was overexpressed by the γ -neural driver, GAL4-201Y [1] (Fig 3C–D). Consistent with this observation, we further found that *E93* overexpression driven by GAL4-OK107 (but not *Worniu*-

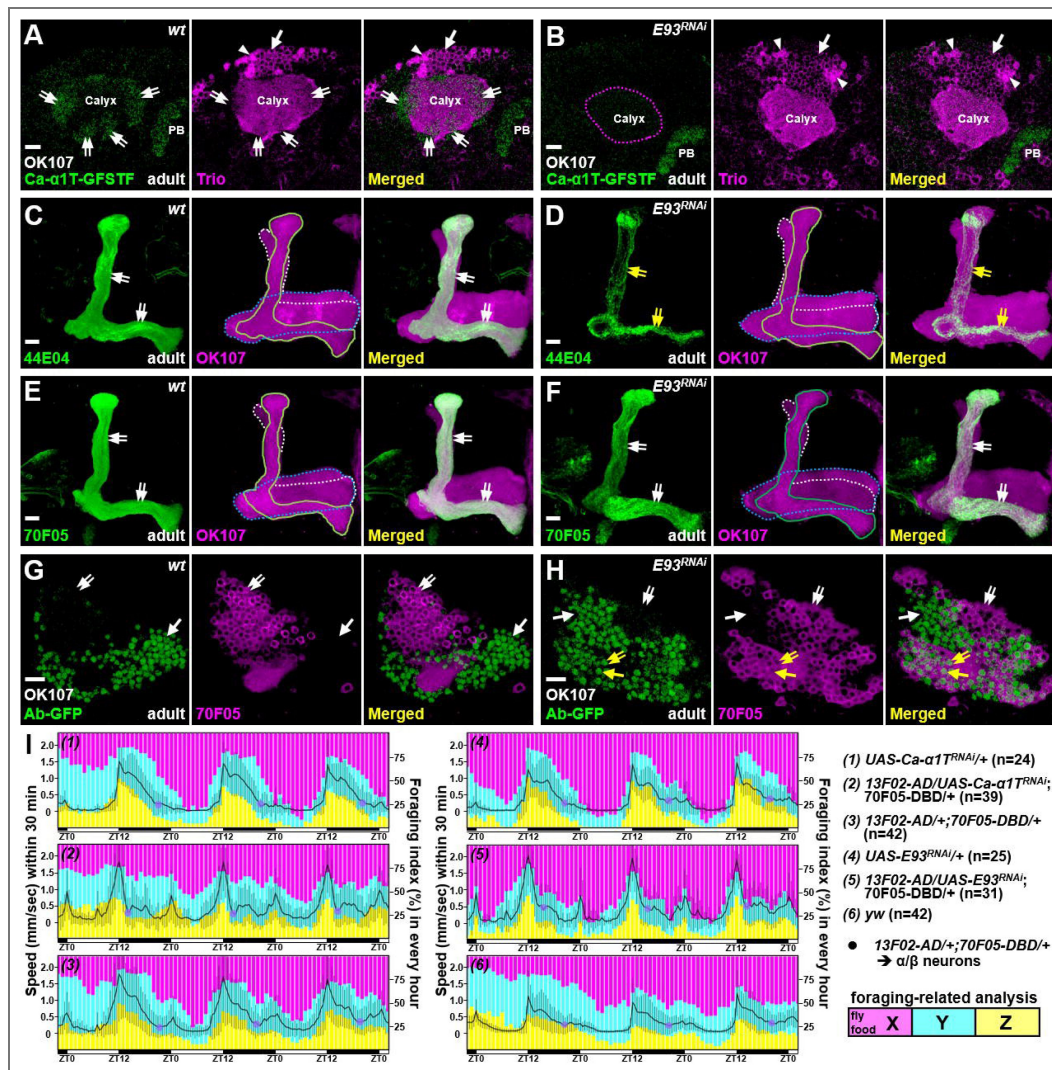


Fig 2. E93 specifies the α/β neural identity and affects animal behaviors

As compared to the wild-type controls (A and C), flies with overexpression of *E93* RNAi (B and D) by a pan-KC driver, GAL4-OK107, had significantly impaired expression of Ca- α 1T-GFSTF and 44E04-LexA in α/β neurons (double arrows) in adult brains. As an internal control, the PB expression of Ca- α 1T-GFSTF was intact under *E93* RNAi knockdown driven by GAL4-OK107. (E-F) *E93* knockdown did not block the expression of 70F05-LexA in α/β neurons to cause the detectable morphological defect in lobe regions (double-arrows). (G-H) However, compared to the wild-type, Ab-GFP expression was ectopically expressed in more than half of 70F05-LexA-positive neurons (double-arrows) when *E93* was knocked down in KCs. The expression levels of 44E04-LexA and 70F05-LexA were visualized by lexAop-myr-GFP in panels C-F and lexAop-mCD8::RFP in panels G and H. Lobes of γ , α/β and α/β neurons were outlined in blue, white and green, respectively, in panels C-F. Cell numbers of 70F05-LexA- and Ab-GFP-positive neurons were counted in S7 Fig. Scale bar: 10 μ m. (I) In control samples, including *yw*, *Ca- α 1T* RNAi-, *E93* RNAi- and α/β neural driver (*13F02-AD/70F05-DBD*)-only flies, it took around eight hours for the minimal speed (purple spots) to be reached at night. However, flies took around two to four hours to achieve a minimal speed when RNAi's for *Ca- α 1T* and *E93* were overexpressed using *13F02-AD/70F05-DBD*. Moving speed (black line) from the second day to the fifth day was calculated as the overall traveling distance (mm) for 30 min. Standard deviation (in grey) for each time-point is shown. The bar graph depicts the duration of food region exploration (X to Z zones, from proximal to distal). All flies, except for *Ca- α 1T* knockdown samples, tended to explore more in the X zone in the day-time. ZT: Zeitgeber time. The setting and analysis of the behavioral assay is detailed in S8 Fig.

GAL4, a pan-neuroblast driver [16]) compromised the expression of specific markers in γ neurons at various developmental stages, including Ab-GFP, the receptor for insect molting hormone ecdysone (EcR-B1) and Mamo isoforms (Fig 3E-J and S9-S10 Fig). Along with the effect of impairing γ -specific marker expression, we further found that E93 overexpression driven by GAL4-OK107 impaired the expression of markers in α/β neurons, including Mamo^{D~G} isoforms and Lac-FSVS (Fig 3I-L). Its overexpression also caused morphological defects by perturbing the remodeling process that normally occurs in early-born KCs at 24-hour after puparium formation (APF; Fig 3K-L). These results taken together suggested that E93 when overexpressed is sufficient to cause specification of KCs toward the α/β neural identity.

Hierarchical genetic network of *chinmo*, *mamo*, *E93* and *ab* controls the cell identity of main KC types

Since Chinmo and E93 act as key TFs to regulate the cell identity of main KC types, we wondered whether *chinmo* and *E93* might form a genetic network to diversify these KC types. Since *chinmo* controls the expression of γ -specific Ab [4, 5] (S2 Fig), we decided to test whether the loss of Ab-GFP by E93 overexpression (seen in Fig 3F) was caused by compromised Chinmo expression. However, we did not observe a reduction of Chinmo level upon E93 overexpression at the first instar larval stage (the most abundant Chinmo expression stage in the wild-type [7]), even though the same manipulation did block Ab-GFP expression (Fig 4A-B). In contrast, E93-GFSTF expression was precociously upregulated in early-born KCs in a *chinmo* mutation line (*chinmo*^[1]) and in the *chinmo* knockdown line (overexpression of *chinmo* RNAi driven by GAL4-OK107) at the WL stage (Fig 4C-D and S11 Fig). In line with the facts that microRNA *let-7* and RNA binding protein Syncrip (Syp) inhibit Chinmo expression [20, 21], we further found that E93-GFSTF expression was abolished by *syp* RNAi knockdown and partially promoted by *let-7* overexpression (S12 Fig). These results suggested that the adoption of γ neural identity by early-born KCs is in part due to suppression of the α/β -neural regulator E93 via Chinmo.

After the diminishment of Chinmo at the early pupal stage [7], we wonder whether E93 would be disinhibited in γ neurons and whether their neural identity would be transformed if this disinhibition indeed occurs? Since γ neurons exist in adult brains and their neural identity is crucially regulated by Mamo at the pupal stage [5], we then tested whether Mamo takes over Chinmo's role to suppress E93 expression in γ neurons. As such, *mamo* RNAi knockdown indeed caused the upregulation of E93-GFSTF in KCs (within enriched Trio expression) of adult brains but not brains at the WL stage (Fig 4E-G), suggesting that Mamo could inhibit E93 expression in γ neurons to ensure their neural identity. Since the E93-mediated α/β neural identity is accompanied by absence of certain γ -specific traits, such as Ab-GFP (Fig 2I and 3F), we next sought to explore whether Ab also plays a crucial role in controlling the KC identity. Intriguingly, we found that the expression of α/β -specific Ca- α 1T-GFSTF and E93-GFSTF was not detectable upon Ab overexpression (Fig 4H-J and S13 Fig). This was accompanied by the expansion of Trio in the cytosol in almost all KCs, indicating the transformation of KCs into γ neuron-like cells. We also noted that RNAi knockdown of *ab* neither abolished Trio expression nor upregulated the expression of Ca- α 1T-GFSTF caused by *E93* knockdown in early-born KCs (S14-S15 Fig). Nonetheless, these results together suggested that *chinmo*, *mamo*, *E93* and *ab* form a hierarchical genetic network with potential feedback loops to control the identity of KCs (Fig 5).

Discussion

Spatially and temporally expressed regulators are employed in the developing nervous system to generate a wide diversity of neuronal types. Spatial patterning cues, such as homeodomain-containing Hox proteins and Sonic Hedgehog, are well documented in the specification of motor neurons and interneurons in the spinal cord [22]. Beyond spatial identity, temporal regulation provides another axis of neuronal diversification: distinct neuron types are sequentially generated to form the six-layer laminar organization of the mammalian cerebral cortex, illustrating how layer-specific neurons arise from neural progenitors according to their birth order [23, 24]. Similarly, in the *Drosophila* central nervous system, neural progenitors (neuroblasts) produce

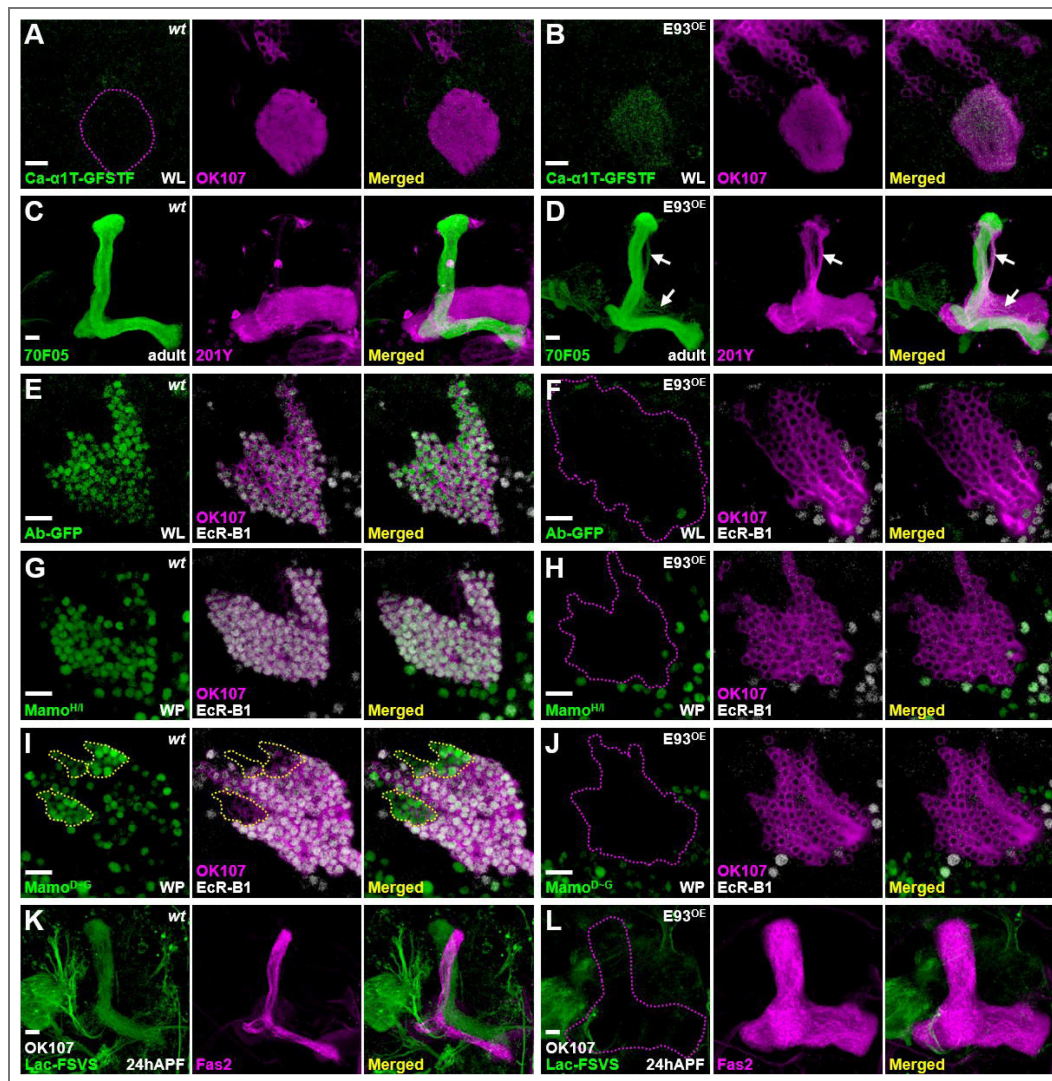


Fig 3. E93 is sufficient to shift the KC identity towards α/β neural-like fate

(A-B) Overexpression of E93 driven by GAL4-OK107 caused precocious expression of α/β -specific Ca- α 1T-GFSTF in early-born KCs at the WL stage. (C-D) In addition, overexpression of E93 driven by a γ -neural driver, GAL4-201Y (magenta), ectopically turned on the expression of a α/β -specific 70F05-LexA driver in a portion of γ neurons (visualized by myr-GFP in green; arrow). (E-J) On the other hand, overexpression of E93 abolished γ -specific markers, including Ab-GFP (E-F), Mamo^{H/I} (G-H), Mamo^{D~G} (weak green signal; I-J) and EcR-B1 (E-J), and α'/β' -specific Mamo^{D~G} (strong green signal within yellow dashed-line; I-J) in the early-born KCs at the white pupal (WP) stage. (K-L) E93 overexpression also compromised the Lac-FSVS expression in α'/β' neurons and the morphology of MB lobes revealed by cell adhesion molecule Fasciclin II (Fas2, strong magenta for labeling α and β lobes) at 24 h after puparium formation (APF). An enhance-promoter (EP) line inserted at the proximal region of the *E93-A* 5'UTR was used to overexpress E93 in the gain-of-function experiments. The potency of the E93(EP) line was similar to two other in-house transgenic lines expressing E93-A and E93-B isoforms (see S9 Fig). Scale bar: 10 μ m.

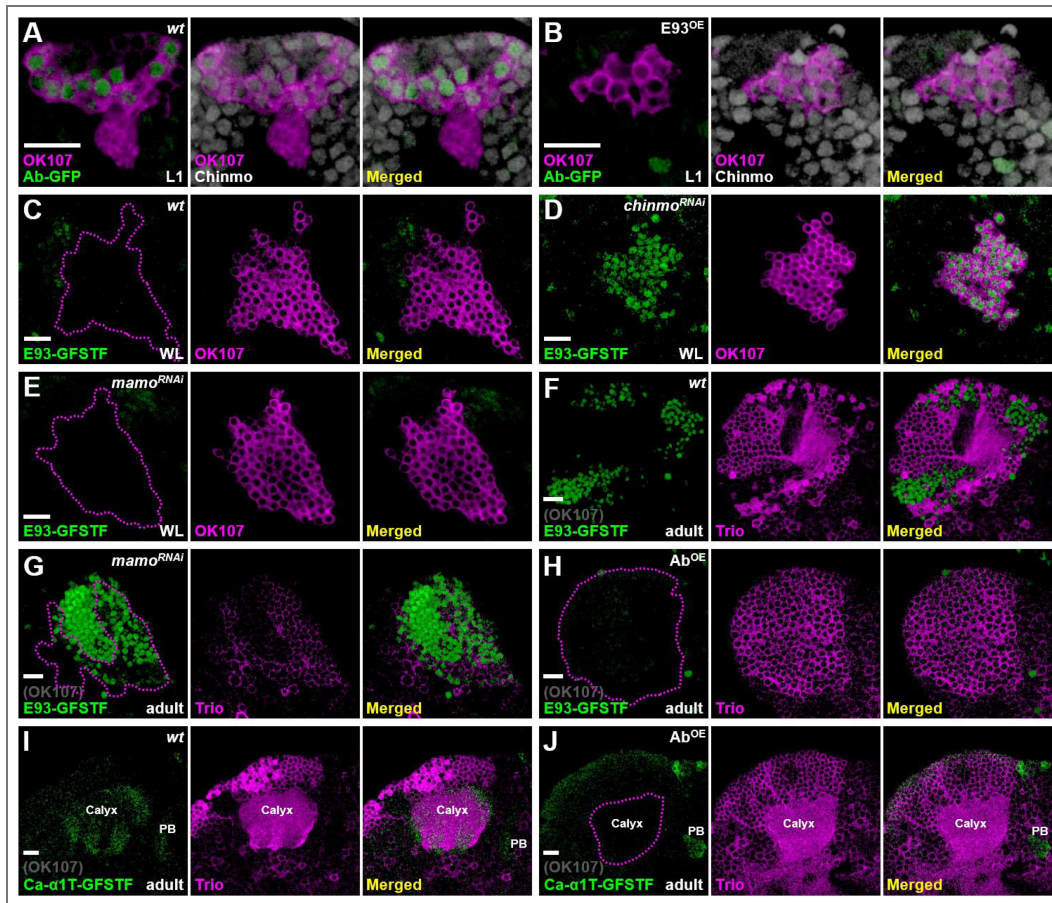


Fig 4. Genetic networks of *chinmo*, *mamo*, E93 and *ab* control KC identity

(A-B) As compared to the wild-type, expression of Ab-GFP (green) was diminished in KCs at the first instar larval (L1) stage upon E93 overexpression driven by GAL4-OK107 (magenta). However, *Chinmo* expression (white) was not affected by E93 overexpression. (C-E) In contrast, RNAi knockdown of *chinmo*, but not *mamo*, driven by GAL4-OK107 (magenta) precociously turned on the expression of E93-GFSTF (green) in the early-born KCs at the WL stage. (F-G) However, RNAi knockdown of *mamo* driven by GAL4-OK107 ectopically turned on expression of E93-GFSTF (green) in KCs with weak cytosolically expressed Trio (magenta) in adult brains (magenta dash-lines). The weak Trio signal was possibly due to *mamo* RNAi knockdown in early-born KCs. E93-GFSTF was densely expressed in putative α/β neurons with negative Trio signal (region outside magenta dashed lines). (H-J) Ab overexpression driven by GAL4-OK107 diminished the expression of E93-GFSTF and Ca- α 1T-GFSTF in KCs of adult brains. Trio seemed to be expressed in the cytosol in almost all KCs upon Ab overexpression. Scale bar: 10 μ m.

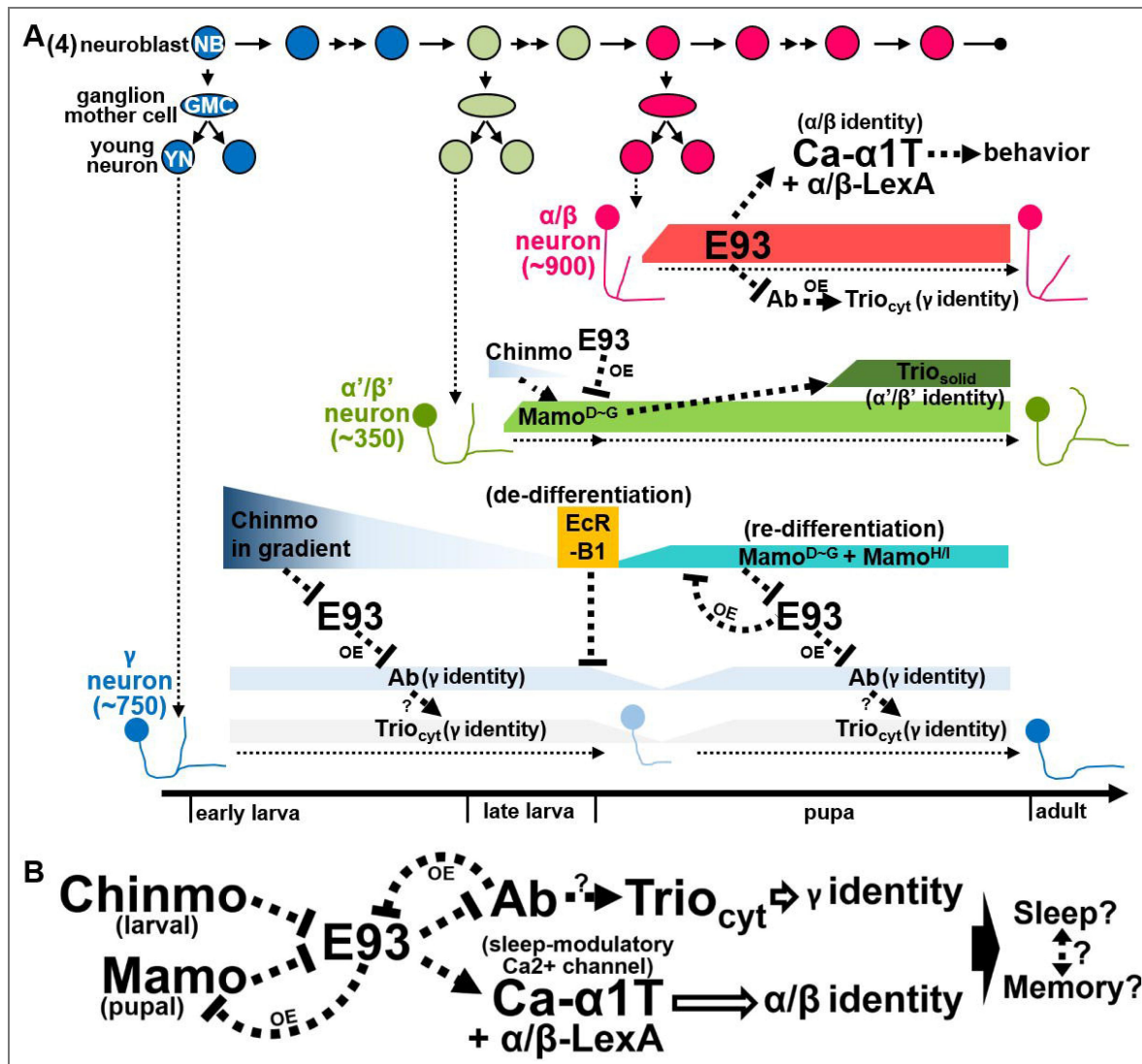


Fig 5. Hierarchical genetic networks govern the identity and function of Kenyon cell in the construction of MBs

(A-B) Scheme delineates hierarchical genetic networks among *chinmo*, *mamo*, *E93* and *ab* with feedback loops that control the cell identities of γ and α'/β' neurons. *Syp* and *let-7* are included in genetic networks to potentially link the regulation of *E93* and sleeping modulatory calcium channel *Ca-α1T* [20, 21, 33] (S12 Fig). Based on the results of gain-of-function studies (Fig 3H–3J and 4H–4J), possible feedback regulation in the genetic network is indicated with OE (as the abbreviation of overexpression). Since *E93* regulates the *Ca-α1T* expression (Fig 2B) and since *let-7* is also crucial for the sleep behavior [33], *E93* and *Ca-α1T* may be potentially associated with sleep and memory behaviors through KCs. Question marks (?) indicate possible regulation in the genetic network.

different neuron types in an invariant sequence under the control of temporal regulators [23, 24]. Distinct sets of transcription factors expressed sequentially in neuroblasts of the embryonic ventral nerve cord and developing medulla direct fate specification, thereby generating diverse neurons for larval and visual circuit assembly [23, 25]. In addition to sequential neuroblast regulation, temporal expression of transcription factors in postmitotic neurons also contributes to neuronal diversity by conferring distinct identities [7, 26]. For example, among the three sequentially generated KC types, γ , α'/β' , and α/β neurons, BTBzf TFs, Chinmo and Mamo, have been shown to play crucial roles in specifying early-born KCs as γ and α'/β' neurons [1, 4, 7]. In this study, we revealed the molecular mechanisms of KC specification by identifying and characterizing Pipsqueak domain-containing TF E93 as a driver of KC identity towards α/β neurons. Our findings that E93 regulates the expression of calcium channel $Ca-\alpha 1T$ to subsequently control animal behaviors also provide an illustration that neural identity specification is associated with acquisition of specific functions among main KC types. We further showed that the specification of KC types is controlled via a genetic network formed by *chinmo*, *mamo*, *E93* and *ab*, which mainly controls the KC identity of γ and α/β neurons during construction of the adult MB (Fig 5C).

Despite that reciprocal regulation between BTBzf TFs and E93 has been reported in multiple cell types during the *Drosophila* development [27, 28], here, we disclosed a novel strategy of using genetic networks of BTBzf TFs and E93 to specify distinct cell identities among neurons derived from the same neuroblasts (Fig 5C). To diversify KC types, Chinmo and Mamo take turns to inhibit the E93 expression, thereby establishing the cell identity of γ neurons, as suggested by Ab expression [4] (Fig 2I and 4G). Without the inhibitory effects of Chinmo and Mamo on E93 expression, late-born KCs specify into α/β neurons by turning on the expression of α/β -specific genes and suppressing the expression of γ -specific genes, Ab included (Fig 2C). Since E93 is an ecdysone-induced protein and Mamo expression is primed by ecdysone signaling in γ neurons [5, 14], blocking EcR-B1 is expected to enhance E93 expression in these neurons. By the fact that overexpression of E93 inhibited EcR-B1 expression in γ neurons (Fig 3F, H and J), future investigations should be able to unravel the intricate regulatory interplay between E93 and ecdysone signaling. In addition, since E93 knockdown did not alter the lobe morphology of α/β neurons (Fig 2F), it is also possible that unidentified regulators might work in concert with E93 to specify the cell fate of α/β neurons. Although how the genetic network of BTBzf TFs and E93 specifies α'/β' neural identity has not yet been fully resolved, the Mamo expression promoted by the low level of Chinmo, which is regulated by the transforming growth factor beta (TGF- β) signaling in MB neuroblasts, is crucial for KCs to adopt the α'/β' neural identity [4, 29]. Intriguingly, our results revealed that E93 overexpression can abolish the expression of Mamo to block KCs from adopting the α'/β' neural identity (Fig 3J and 3L). All these results taken together portray that the genetic network of BTBzf TFs and E93 is very likely to control the KC identity during construction of the adult MB. In light of the phylogenetic conservation of the E93 family and *tramtrack* (TTK)-type BTBzf TFs, including *chinmo*, *mamo* and *ab*, as Arthropoda-specific genes [30, 31], one cannot help but wonder whether TTK-type BTBzf TFs might have been evolutionally introduced to intervene with E93 to regulate the development of multiple arthropodan tissues; perhaps future investigations may delineate their relationship on neuron-type diversification in the nervous system.

Once the identities of neuron types are determined by key regulators, how the cells acquire specific functions in the nervous system to eventually participate in animal behaviors? Previous findings together with our current studies might shed some light on this topic. First, the expression of Chinmo and Ab is inhibited in late-born KCs, in part due to the preferential expression of *let-7* in α/β neurons [21, 32]. In addition, the neuronal (and KC) contribution of *let-7* regulates day- and night-time sleeping behaviors in developmental- and adult-restricted manners [33]. Furthermore, Chinmo and *let-7* potentially regulate the expression of E93, for which expression in α/β neurons governs the expression of a phylogenetically conserved sleep-modulatory calcium channel $Ca-\alpha 1T$ to regulate animal behaviors [18] (Fig 2B, 4D and S12A Fig). Interestingly, blocking neurotransmission with tetanus toxin light chain (TeNT-Ln) in the MB has been shown to promote locomotion, an outcome opposite to the *Ca-\alpha 1T* knockdown results observed in our study [34] (Fig

2I and S8E Fig). This discrepancy may stem from TeNT-Ln acting on MB axons to inhibit downstream neurons, whereas the effect of *Ca- α 1T* knockdown is primarily occurred MB dendrites that receive inputs from upstream neurons [34] (Fig 1H). These results together with the previous reported function of E93 in circadian rhythm provide an intriguing link between E93 and sleep-associated behaviors [35]. Since KC types were reported to function crucially in sleep/arousal and alcohol-induced sleep deficit behaviors [36–38], our finding of KC identity specification and *Ca- α 1T* expression under the regulation of E93 suggests that this process may in some way endow α/β neurons with the capacity to control animal behaviors related to sleep. Since the intricate relationship between sleep and memory is well established [39–42], future deeper investigations on cell-type specification of memory-associated neurons are expected to provide insights into how to acquire for unique functions among neurons to regulate these two crucial traits of animals.

Materials and Methods

Experimental model and subject details

Flies were cultured in a room maintained at 25°C ($\pm$ 1.5°C) and 50–65% humidity for all experiments. For most experiments, flies were used with no selection for sex; therefore, roughly equal numbers of males and females were used. However, since homozygous *Ca- α 1T-GFSTF* females and hemizygous *Ca- α 1T-GFSTF* males are relatively feeble, heterozygous *Ca- α 1T-GFSTF* females were selected for the analysis in Fig. 1G–H, 2A–B, 3A–B and 4I–J and S5A–B, S13 and S14D Figs due to the cytolocation of the *Ca- α 1T* gene on the X chromosome. In addition, males were selected for the behavioral assay to mitigate complications of egg laying behaviors on locomotion tracking.

Fly strains

The fly strains used in this study were as follows. Most strains are available from either Bloomington *Drosophila* stock center (BDSC), Kyoto *Drosophila* stock center (DGGR) or Vienna *Drosophila* stock center (VDRC). (1) *Ab-GFP* (BDSC38626); (2) *Lac-FSVS* (DGGR115308); (3) *E93-GFSTF* (BDSC59412); (4) *Ca- α 1T-GFSTF* (BDSC61800); (5) *hs-FLP¹²,UAS-mCD8::GFP* (BDSC28832); (6) *tubP-GAL80,FRT^{40A}* (BDSC5192); (7) *chinmo¹,FRT^{40A}* (BDSC59969); (8) *GAL4-OK107* (BDSC854); (9) *UAS-mCD8::RFP [attp40]* (BDSC32219); (10) *UAS-mCD8::RFP [attp2]* (BDSC32218); (11) *UAS-E93 RNAi* (BDSC57868); (12) *UAS-E93 RNAi* (VDRC104390); (13) *UAS-Ca- α 1T RNAi* (BDSC39029); (14) *GAL4-c739* (BDSC7362); (15) *44E04-LexA::P65* (BDSC52736); (16) *70F05-LexA::P65* (BDSC523629); (17) *lexAop2-myr::GFP* [43]; (18) *lexAop2-mCD8::RFP* [43]; (19) *FRT^{82B},tubP-GAL80* (BDSC5135); (20) *FRT^{82B}* (BDSC86313); (21) *E93 ^{Δ 11}* (BDSC93128); (22) *E93(EP)* (BDSC30179); (23) *UAS-E93-A [VK37]* (this study); (24) *UAS-E93-B [VK37]* (this study); (25) *GAL4-201Y* (BDSC4440); (26) *mamo^{H/I}-HA* [44]; (27) *mamo^{D~G}-HA* [44]; (28) *UAS-chinmo RNAi [VK37]* [5]; (29) *UAS-LUC-let7* (BDSC41171); (30) *UAS-Ssyp RNAi* (VDRC33011); (31) *UAS-mamo RNAi* (BDSC44103); (32) *UAS-ab* (BDSC23639); (33) *UAS-ab-HA* (FlyORF000705); (34) *UAS-Dcr2* (BDSC24651); (35) *UAS-ab RNAi* (VDRC104582); (36) *13F02-p65.AD* (BDSC68291); (37) *70F05-GAL4.DBD* (BDSC69380). The *UAS-E93-A* and *UAS-E93-B* transgenes were generated using standard molecular biology methods to clone cDNA fragments derived from fully-sequenced EST clones, GH10557 and LP08695 (available from *Drosophila* Genomics Resource Center), carrying *E93-A* and *E93-B* isoforms into the attB-FAST vector. The generation of *UAS-E93-A* and *UAS-E93-B* transgenes and fly stains was performed by WellGenetics, Inc.

RNAi knockdown and overexpression experiments and MARCM clonal analyses

UAS-RNAi and *UAS-transgene* lines were crossed to *GAL4-107* and *GAL4-201Y* for knockdown and overexpression of genes of interest in KCs. Mosaic clones for the MARCM studies were generated as previously described [45]. In short, mosaic clones of *chinmo¹* and *E93 ^{Δ 11}* mutations were induced by 35 min of heat shock using *hs-FLP¹²* in newly hatched larva. Dissection, immunostaining and mounting of adult brains were performed as described in a standard

protocol [45]. Primary antibodies used in this study included guinea pig antibody against Chinmo (1:1000, Sokol laboratory [46]), rat monoclonal antibody against mCD8 (1:100, Thermo Fisher Scientific), rabbit antibody against GFP (1:750, Thermo Fisher Scientific), and mouse monoclonal antibodies against EcR-B1 (1:50, DSHB), Fas2 (1:100, DSHB) and Trio (1:50, DSHB). Secondary antibodies conjugated to different fluorophores (Alexa 488, Alexa 546 and Alexa 647; Thermo Fisher Scientific and Jackson ImmunoResearch Lab, Inc.) were used at 1:750 dilutions. Immunofluorescence images were collected by confocal microscopy on a Zeiss LSM 700, projected using the LSM browser and processed in Adobe Photoshop CS6. All data are representative of more than 3 brains per genotype.

Distance measurement in the behavioral assay and statistical analysis

Individual fly activities were recorded and analyzed using the activity monitor system developed by DroBot, Inc. This system used an open-resourced software pySolo to track individual moving flies [47]. Average speed and standard deviation of individual flies were calculated according to total traveling distance within 30 min periods from the first day to the fifth day, as shown in Fig 2I and S8E Fig. Student's t-test was used for statistical analysis to compare datasets with two groups in S8 Fig.

Data availability

Microscopy data reported in this paper will be shared by the lead contact upon request. Any additional information required to reanalyze the data reported in this paper is available from the corresponding author upon request.

Acknowledgements

We thank the TRiP at Harvard Medical School (NIH/NIGMS R01-GM084947) for providing the transgenic RNAi fly stocks used in this study. We also thank DroBot Inc. for designing the behavioral assay system used in this study. This work was supported by National Science and Technology Council (NSTC-112-2311-B-001-029) and Thematic Research Program of Academia Sinica (AS-TP-113-L01), Taiwan.

Additional information

Author contributions

Conceptualization, P.C.C. and H.H.Y.; Methodology, P.C.C., K.Y.K., S.Y.H. C.C., and H.H.Y.; Formal Analysis, P.C.C., C.C. and H.H.Y.; Investigation, P.C.C. and H.H.Y.; Writing: P.C.C. and H.H.Y.; Visualization, P.C.C., C.C. and H.H.Y.; Supervision: H.H.Y., Funding Acquisition, H.H.Y.

Materials and correspondence

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Additional files

[Supplementary figures and tables](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

The temporal regulation of neuronal specification and its molecular mechanisms are important problems in developmental neurobiology. This study focuses on Kenyon cells (KCs), which form the mushroom body in *Drosophila melanogaster*, in order to address this issue. Building on previous findings, the authors examine the role of the transcription factor Eip93F in the development of late-born KCs. The authors revealed that Eip93F controls the activity of flies at night through the expression of the calcium channel Ca- α 1T. Thus, the study clarifies the molecular machinery that controls temporal neuronal specification and animal behavior.

Strengths:

The convincing results are based on state-of-the-art molecular genetics, imaging, and behavioral analysis.

<https://doi.org/10.7554/eLife.108173.2.sa2>

Reviewer #2 (Public review):

Summary:

Understanding the mechanisms of neural specification is a central question in neurobiology. In *Drosophila*, the mushroom body (MB), which is the associative learning region in the brain, consists of three major cell types: γ , α'/β' and α/β kenyon cells. These classes can be further subdivided into seven subtypes, together comprising ~2000 KCs per hemi-brain. Remarkably, all of these neurons are derived from just four neuroblasts in each hemisphere. Therefore, a lot of endeavours are put to understand how the neuron is specified in the fly MB.

Over the past decade, studies have revealed that MB neuroblasts employ a temporal patterning mechanism, producing distinct neuronal types at different developmental stages. Temporal identity is conveyed through transcription factor expression in KCs. High levels of Chinmo, a BTB-zinc finger transcription factor, promote γ -cell fate (Zhu et al., Cell, 2006). Reduced Chinmo levels trigger expression of mamo, a zinc finger transcription factor that specifies α'/β' identity (Liu et al., eLife, 2019). However, the specification of α/β neurons remains poorly understood. Some evidence suggests that microRNAs regulate the transition from α'/β' to α/β fate (Wu et al., Dev Cell, 2012; Kucherenko et al., EMBO J, 2012). One hypothesis even proposes that α/β represents a "default" state of MB neurons, which could explain the difficulty in identifying dedicated regulators.

The study by Chung et al. challenges this hypothesis. By leveraging previously published RNA-seq datasets (Shih et al., 2019), they systematically screened BAC transgenic lines to selectively label MB subtypes. Using these tools, they analyzed the consequences of manipulating E93 expression and found that E93 is required for α/β specification. Furthermore, loss of E93 impairs MB-dependent behaviors, highlighting its functional importance.

Strengths:

The authors conducted a thorough analysis of E93 manipulation phenotypes using LexA tools generated from the Janelia Farm and Bloomington collections. They demonstrated that E93 knockdown reduces expression of Ca- α 1T, a calcium channel gene identified as an α/β marker. Supporting this conclusion, one LexA line driven by a DNA fragment near EcR (R44E04) showed consistent results. Conversely, overexpression of E93 in γ and α'/β' Kenyon cells led to downregulation of their respective subtype markers.

Another notable strength is the authors' effort to dissect the genetic epistasis between E93 and previously known regulators. Through MARCM and reporter analyses, they showed that Chinmo and Mamo suppress E93, while E93 itself suppresses mamo. This work establishes a compelling molecular model for the regulatory network underlying MB cell-type specification.

Weaknesses:

The interpretation of E93's role in neuronal specification requires caution. Typically, two criteria are used to establish whether a gene directs neuronal identity:

- (1) gene manipulation shifts the neuronal transcriptome from one subtype to another, and
- (2) gene manipulation alters axonal projection patterns.

The results presented here only partially satisfy the first criterion. Although markers are affected, it remains possible that the reporter lines and subtype markers used are direct transcriptional targets of E93 in α/β neurons, rather than reflecting broader fate changes. Future studies using transcriptomics would provide a more comprehensive assessment of neuronal identity following E93 perturbation.

With respect to the second criterion, the evidence is also incomplete. While reporter patterns were altered, the overall morphology of the α/β lobes appeared largely intact after E93 knockdown. Overexpression of E93 in γ neurons produced a small subset of cells with α/β -like projections, but this effect warrants deeper characterization before firm conclusions can be drawn.

Overall, this study has nicely shown that E93 can regulate α/β neural identities. Further studies on the regulatory network will help to better understand the mechanism of neurogenesis in mushroom body.

<https://doi.org/10.7554/eLife.108173.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

*The temporal regulation of neuronal specification and its molecular mechanisms are important problems in developmental neurobiology. This study focuses on Kenyon cells (KCs), which form the mushroom body in *Drosophila melanogaster*, in order to address this issue. Building on previous findings, the authors examine the role of the transcription factor Eip93F in the development of late-born KCs. The authors revealed that Eip93F controls the activity of flies at night through the expression of the calcium channel Ca- α 1T. Thus, the study clarifies the molecular machinery that controls temporal neuronal specification and animal behavior.*

Strengths:

The convincing results are based on state-of-the-art molecular genetics, imaging, and behavioral analysis.

Weaknesses:

Temporal mechanisms of neuronal specification are found in many nervous systems. However, the relationship between the temporal mechanisms identified in this study and those in other systems remains unclear.

We have discussed the temporal mechanisms between different nervous systems at the beginning of the Discussion section.

Reviewer #2 (Public review):

Summary:

*Understanding the mechanisms of neural specification is a central question in neurobiology. In *Drosophila*, the mushroom body (MB), which is the associative learning region in the brain, consists of three major cell types: γ , α'/β' , and α/β kenyon cells. These classes can be further subdivided into seven subtypes, together comprising ~2000 KCs per hemi-brain. Remarkably, all of these neurons are derived from just four neuroblasts in each hemisphere. Therefore, a lot of endeavors are put into understanding how the neuron is specified in the fly MB.*

Over the past decade, studies have revealed that MB neuroblasts employ a temporal patterning mechanism, producing distinct neuronal types at different developmental stages. Temporal identity is conveyed through transcription factor expression in KCs. High levels of Chinmo, a BTB-zinc finger transcription factor, promote γ -cell fate (Zhu et al., Cell, 2006). Reduced Chinmo levels trigger expression of mamo, a zinc finger transcription factor that specifies α'/β' identity (Liu et al., eLife, 2019). However, the specification of α/β neurons remains poorly understood. Some evidence suggests that microRNAs regulate the transition from α'/β' to α/β fate (Wu et al., Dev Cell, 2012; Kucherenko et al., EMBO J, 2012). One hypothesis even proposes that α/β represents a "default" state of MB neurons, which could explain the difficulty in identifying dedicated regulators.

The study by Chung et al. challenges this hypothesis. By leveraging previously published RNA-seq datasets (Shih et al., G3, 2019), they systematically screened BAC transgenic lines to selectively label MB subtypes. Using these tools, they analyzed the consequences of manipulating E93 expression and found that E93 is required for α/β specification. Furthermore, loss of E93 impairs MB-dependent behaviors, highlighting its functional importance.

Strengths:

The authors conducted a thorough analysis of E93 manipulation phenotypes using LexA tools generated from the Janelia Farm and Bloomington collections. They demonstrated that E93 knockdown reduces expression of $Ca_{\alpha 1T}$, a calcium channel gene identified as an α/β marker. Supporting this conclusion, one LexA line driven by a DNA fragment near EcR (R44E04) showed consistent results. Conversely, overexpression of E93 in γ and α'/β' Kenyon cells led to downregulation of their respective subtype markers.

Another notable strength is the authors' effort to dissect the genetic epistasis between E93 and previously known regulators. Through MARCM and reporter analyses, they showed that Chinmo and Mamo suppress E93, while E93 itself suppresses Mamo. This work establishes a compelling molecular model for the regulatory network underlying MB cell-type specification.

Weaknesses:

The interpretation of E93's role in neuronal specification requires caution. Typically, two criteria are used to establish whether a gene directs neuronal identity:

(1) gene manipulation shifts the neuronal transcriptome from one subtype to another, and

(2) gene manipulation alters axonal projection patterns.

The results presented here only partially satisfy the first criterion. Although markers are affected, it remains possible that the reporter lines and subtype markers used are direct transcriptional targets of E93 in α/β neurons, rather than reflecting broader fate changes. Future studies using single-cell transcriptomics would provide a more comprehensive assessment of neuronal identity following E93 perturbation.

We do plan to conduct multi-omics experiments to provide a more comprehensive assessment of neuronal identity upon loss-of-function of E93. However, omics results take time to be conducted and analyzed, so the result will be summarized in a future manuscript.

With respect to the second criterion, the evidence is also incomplete. While reporter patterns were altered, the overall morphology of the α/β lobes appeared largely intact after E93 knockdown. Overexpression of E93 in γ neurons produced a small subset of cells with α/β -like projections, but this effect warrants deeper characterization before firm conclusions can be drawn. While the results might be an intrinsic nature of KC types in flies, the interpretation of the reader of the data should be more careful, and the authors should also mention this in their main text.

We have toned down our description on the effect of E93 (especially in the loss-of-function) in specifying the α/β -specific cell identity and discussed whether unidentified regulators would work together with E93 in α/β neural fate specification.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Changes in nighttime activity in flies upon knocking down $Ca_{\alpha 1T}$ and Eip93F are interesting (Fig. 2C). However, examining the morphological changes in the mushroom body under these conditions would be essential.

We did not find the morphological change of mushroom body lobes by examining with the Fas2 staining (shown in Figure S8D).

(2) Temporal mechanisms of neuronal specification have been identified in various nervous systems, including the embryonic central nervous system (CNS), the optic lobe of

Drosophila, and the nervous systems of other organisms. The Discussion section should address the relationship between the temporal mechanisms identified in this study and those identified in other systems.

We have discussed the temporal mechanisms between different nervous systems at the beginning of the Discussion section.

(3) Eip93F is an Ecdysone-induced protein. In the Discussion section, the authors should discuss the relationship between the ecdysone signal and the roles of Eip93F.

We have added the discussion on the relationship between the ecdysone signal and the roles of Eip93F.

Reviewer #2 (Recommendations for the authors):

(1) The behavioral effect of Ca- α 1T knockdown is pretty interesting. But how the downregulation of Ca- α 1T in the mushroom body can affect locomotion is puzzling. Even though the mushroom body is known to suppress locomotion (Matin et al., Learn Mem, 1998), the real results are opposite. Can authors give further explanation in the discussion? Also, the behavioral experiments are hard to interpret, given that Figure 2C(1) and Figure 2C(3) as a control, also vary a lot. Since the behavioral experiments don't affect the main conclusion of the paper, I would suggest removing that part or adding more explanation in the discussion.

First, we have discussed the puzzling part on the MB influence in locomotion between the previous study using tetanus toxin light chain (TeNT-Ln) and our Ca- α 1T knockdown result. It is possible that the different effect is derived from TeNT-Ln's function in MB axons and Ca- α 1T's function in MB dendrites. Secondly, we have re-conducted the behavioral results using a new α/β driver (13F02-AD/70F05-DBD) to replace our initial behavioral results (using c739-GAL4, which would cause the abnormal wing when drives E93 RNAi expression; see S8C(2) Fig). Current results (now in Fig 2I) are more consistent in control groups.

(2) In the manuscript, the authors use "subtype" to describe γ KC, α'/β' KC and α/β KC in the fly MB. However, in most of the literature, people use "main types" to summarize these three types, and "subtype" is mostly about the difference in γ d, γ m, α'/β' ap, α'/β' m, α/β p, α/β s and α/β c KC (Shih et al., G3, 2019). Replacing "subtypes" with "main types" will help to increase the clarity.

We have replaced "KC subtypes" with "main KC types" or just "KC types".

(3) The authors have identified a lot of new markers for the KC cell types, and some of them are used in this manuscript. It will be helpful if they can have a figure to summarize the markers they used in this study and what cell types they labeled.

We have summarized expression patterns of these markers in Supplemental table 1.

(4) In the method, the authors mentioned that only females were selected for analysis of Ca- α 1T-GFSTF. Could the authors explain the reasons in more detail?

Since homozygous Ca- α 1T-GFSTF female flies and hemizygous Ca- α 1T-GFSTF male are a bit sick and hard to collect, we therefore used heterozygous Ca- α 1T-GFSTF female in our experiments. I have added this description in the Materials and Methods section.

(5) Figure S1: The legend of magenta fluorescence is missing. Please add which protein is shown in magenta.

We have added the legend of magenta fluorescence, which is Trio.

(6) The detailed genotypes of Figure 2C and Figure S7 are missing in Supplementary Table 1. Please include that, so that readers can know the genetic background.

We have added genotypes of Figure 2I (previously Figure 2C) and Figure S8 (previously as Figure S7) in Supplementary Table 2.

(7) Figure 2D-G: It will be helpful if the authors can outline the lobe (γ , α'/β' , and α/β) in the figure, which will help readers to understand the images.

We have outlined α , α' , β , β' and γ lobes in Figure 2C-F (previously as Figure 2D-G).

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